De La Salle Holy Cross College

Grade 10 Mathematics November Examination – Paper 2

Examiner: Mathematics Department Total: 100

Moderators: Grade 10 Mathematics Teachers Time: 2 hours

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Instructions and Information:

- This question paper consists of 18 pages which include 10 questions.
- Page 18 is blank and may be used for additional working. Clearly indicate the question number should you use this extra space.
- Answer all the questions in the space provided.
- You may use any permitted calculator. Ensure that it is set in degree mode.
- All necessary working-out must be shown clearly; providing only final answers does not necessarily guarantee full marks.
- Diagrams and graphs are not necessarily drawn to scale.
- If necessary, round off answers to **ONE** decimal place, unless otherwise stated.
- Write neatly and legibly; anything that is unclear will not be marked.

USEFUL FORMULAE

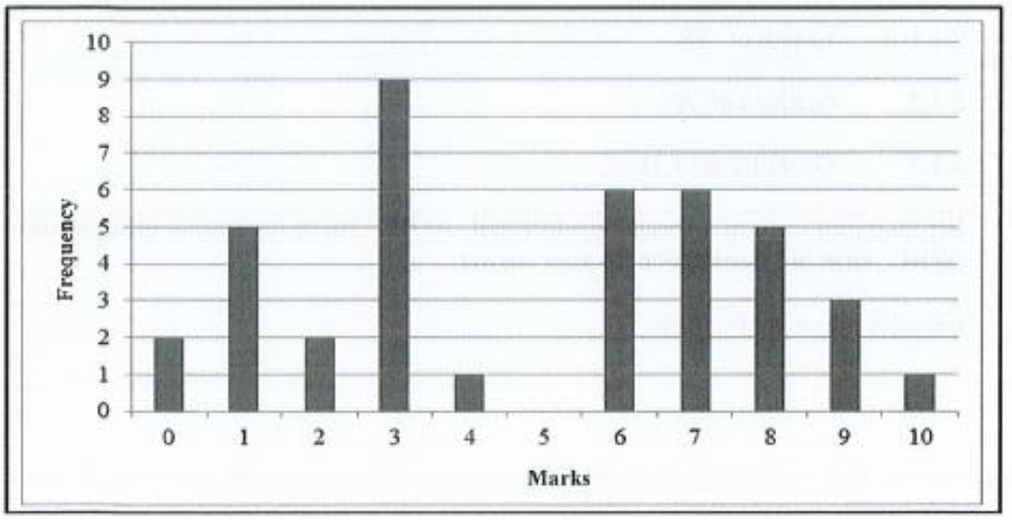
$\left(\frac{x_1+x_2}{2};\frac{y_1+y_2}{2}\right)$ $m=\frac{y_2-y_1}{x_2-x_1}$ $y=mx+c$ $d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

QU	Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	Q 9	Q 10	Total
Mark											
Total	10	10	10	7	13	15	8	8	9	10	100
Marker											

QUESTION 1 [10 Marks]

Grade 10 learners wrote a class test that had a maximum mark of 10. The teacher represented the marks for the class in the bar graph shown below.

Bar graph to show the distribution of marks out of 10



(a) How many of the learners scored at least 8 out of 10 for the test? (2)

9 ✓

(b) How many learners are in this class? (1)

40 ✓

(c) Calculate the range of the marks. (2)

10 − 0 ✓ = 10 ✓

(d) If the pass mark for the test is 50%, determine the percentage of learners who failed the test. (2)

✓ $\frac{19}{40}$ × 100% = 47,5% ✓

(e) Calculate the mean mark for the test. Give your answer correct to **3** decimal places. (3)

✓ $\frac{195}{40}$ ✓ = 4,875 ✓

(Can get 3 if answer only and it is correct)

QUESTION 2 [10 Marks]

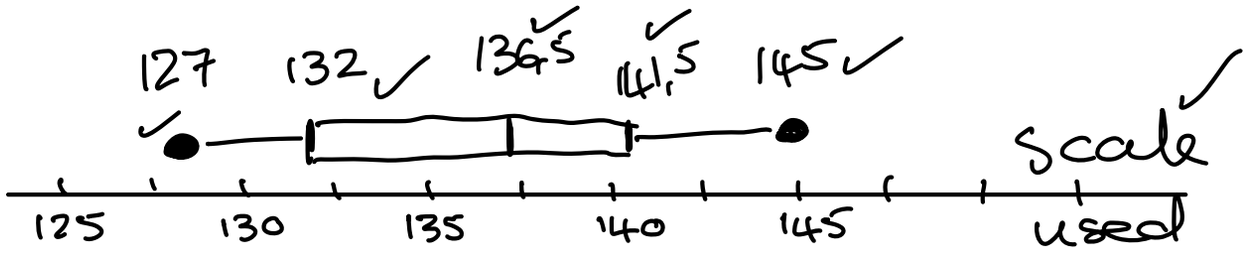
The heights (in centimetres) of 20 children were measured and recorded in the table below.

127	128	129	130	131	133	134	134	135	136
137	138	139	140	141	142	142	143	144	145

(a) Calculate the interquartile range for the heights of the children. (3)

$$141,5 - 132 = 9,5$$

(b) Draw a box and whisker diagram to represent the data shown in the table. (6)



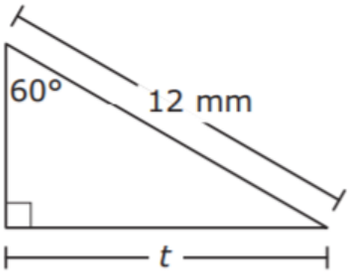
(c) Use the information in your box and whisker diagram to comment on the distribution of the heights of the children. Choose the best description from the choices below:

A skewed to the left/negatively skewed
B reasonably symmetrical
C skewed to the right/ positively skewed. (1)

QUESTION 3 [10 Marks]

For each of the questions that follow choose the letter that represents the correct answer. You do not need to show calculations.

(a) A right-angled triangle is drawn. Based on the measurements that are given, what is the length of the side marked t?



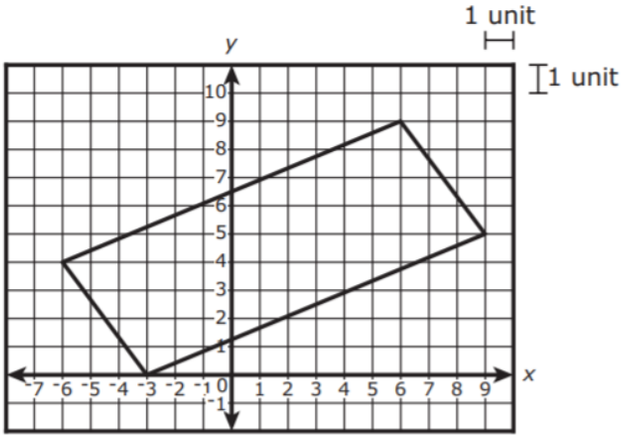
A	$6\ mm$
B	$4\sqrt{3}\ mm$
C	$6\sqrt{3}\ mm$
D	$8\ mm$

✓✓

$\sin 60 = \frac{t}{12}$

$6\sqrt{3} = t$

(b) A parallelogram is plotted on the Cartesian Plane.
What is the perimeter of this parallelogram?



A	24 units
B	36 units
C	48 units
D	64 units

✓✓

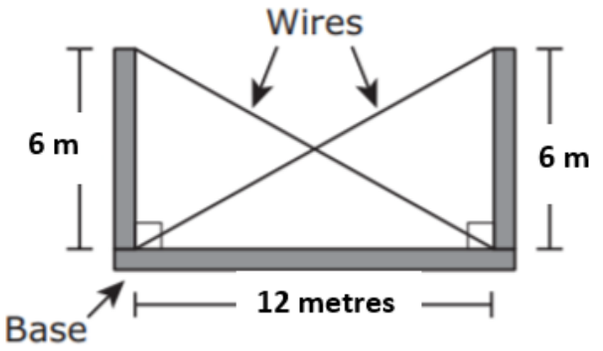
$5 + 5 + 13 + 13$

$= 36$

(c) Gigi has four identical wooden rods. Using them she can form:

A	a square or a rhombus, but not a rectangle.
B	a rhombus, but not a parallelogram.
C	a square or a rectangle, but not a parallelogram.
D	a square, but not a rhombus or a rectangle.
E	a square, rhombus, rectangle or parallelogram.

(d) A large storage rack has two wires that support it. Each wire connects the top of the one side to the base. The rack and its interior dimensions are shown in the diagram. Which of the following is CLOSEST to the length of each of the wires?



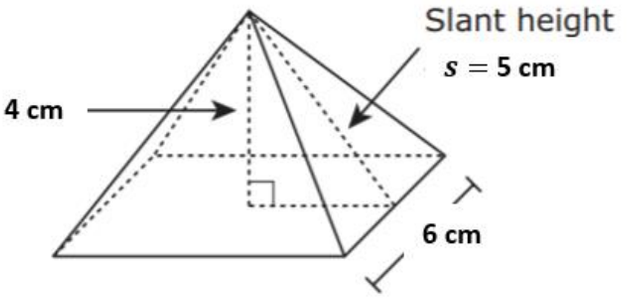
A	9 m
B	10,4 m
C	13,4 m
D	18 m

$6^2 + 12^2 = l^2$

$\sqrt{180}$

(e) The diagram shows a square based pyramid and some of its dimensions. What is the volume of the pyramid?

$Volume = \frac{1}{3} Area\ of\ base \times Height$



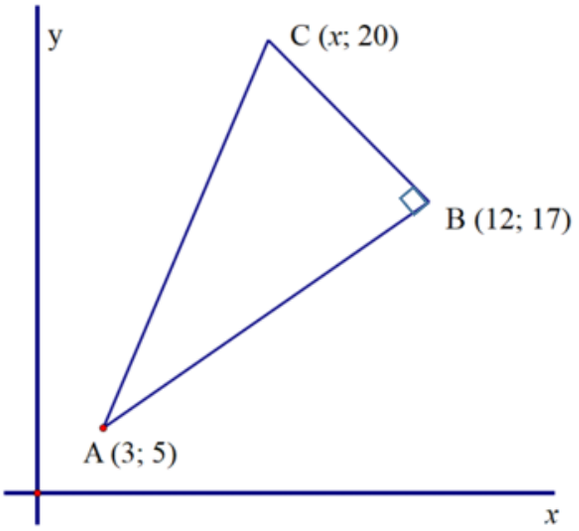
A	48 cm^3
B	60 cm^3
C	144 cm^3
D	180 cm^3

$\frac{1}{3} \times 36 \times 4$

$= 48$

QUESTION 4 [7 Marks]

In the diagram below ΔABC is a right-angled scalene triangle with AB $\perp$ BC.
The vertices of the triangle are given as $A\ (3; 5)$, $B\ (12; 17)$ and $C\ (x; 20)$.



(a) Determine the gradient of AB. (2)

$$m_{AB} = \frac{17-5}{12-3} = \frac{4}{3} \quad (\text{or } \frac{12}{9})$$

(b) Write down the gradient of BC. (1)

$$m_{BC} = -\frac{3}{4} \quad \text{Ca}$$

(c) Hence, or otherwise, calculate the value of x. (4)

$$\frac{20-17}{x-12} = -\frac{3}{4} \quad \text{Ca}$$

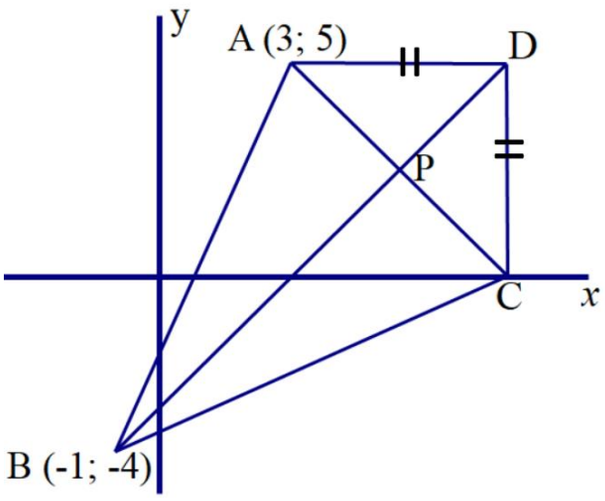
$$3x - \frac{4}{3} = x - 12 \quad \text{m}$$

$$8 = x$$

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QUESTION 5 [13 Marks]

In the diagram below, ABCD is a kite with A $(3;\ 5)$ and B $(-1;\ -4)$.
$AD = DC$ and $AB = BC$.
D is a point such that AD is parallel to the x-axis and $AD = 5$ units.
$CD \perp x$-axis. The diagonals of the kite intersect at P.



(a) Explain briefly why the coordinates of C are (8; 0). (2)

D(8; 5) ✓ so C(8; 0) ✓
same vertically below on x-axis

(b) Calculate the coordinates of P. (2)

$P\left(\frac{3+8}{2}; \frac{5+0}{2}\right)$

$P\left(\frac{11}{2}; \frac{5}{2}\right)$ ✓ ✓ or $(5,5; 2,5)$

(c) Use analytical formulae to prove that AC is perpendicular to BD. (4)

$m_{AC} = \frac{5-0}{3-8} = -1$ ✓

$m_{BD} = \frac{-4-5}{-1-8} = \frac{-9}{-9} = 1$ ✓

-1×1 ✓ $= -1$ ✓

$\therefore AC \perp BD$

(d) Calculate the length of AC. Leave your answer in surd form. (2)

$$AC = \sqrt{(3-8)^2 + (5-0)^2} \checkmark$$

$$= \sqrt{25+25}$$

$$= \sqrt{50}$$

$$= 5\sqrt{2} \checkmark$$

(e) Calculate the area of the kite ABCD. (3)

$$BD = \sqrt{(-1-8)^2 + (-4-5)^2}$$

$$= \sqrt{162} \checkmark = 9\sqrt{2}$$

$$\text{Area} = \frac{1}{2} d_1 \times d_2$$

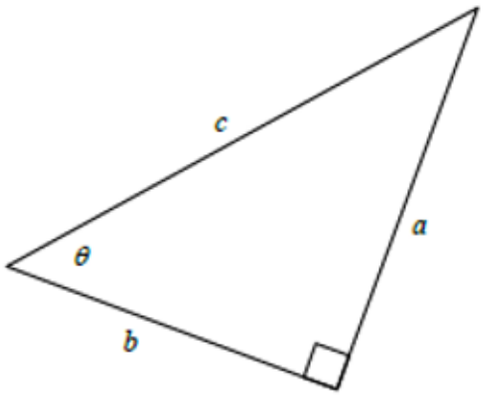
$$= \frac{1}{2} \times \sqrt{50} \times \sqrt{162} \checkmark$$

$$= \frac{1}{2} \times \sqrt{8100}$$

$$= \underline{45 \text{ u}^2} \checkmark$$

QUESTION 6 [15 Marks]

(a) A right-angled triangle has sides a, b, and c and an angle marked θ, as shown in the sketch.



Write down, in terms of a, b, c and θ the ratio for:

(1) $cos\ \theta$ $= \frac{b}{c}$ ✓ (1)

(2) $tan\,\theta$ $= \frac{a}{b}$ ✓ (1)

(3) $sin\,(90° - \theta)$ $= \frac{b}{c}$ ✓ (1)

(b) If $2\cos A = -1$ and $A \in [0°; 180]$, use a diagram but no calculator to determine the values of each of the following:

(1) $sin^2A + cos^2A$

$\left(\frac{\sqrt{3}}{2}\right)^2$ ✓ $+\left(-\frac{1}{2}\right)^2$ ✓

$\frac{3}{4} + \frac{1}{4}$

$= 1$ ✓

Diagram



(5)

(2) $\frac{\tan A}{\sin A}$ $= \frac{\frac{\sqrt{3}}{-1}}{\frac{\sqrt{3}}{2}}$ $= \frac{\sqrt{3}}{-1} \times \frac{2}{\sqrt{3}} = -2$ ✓ ✓ (2)

(c) This question must be done WITHOUT USING A CALCULATOR.

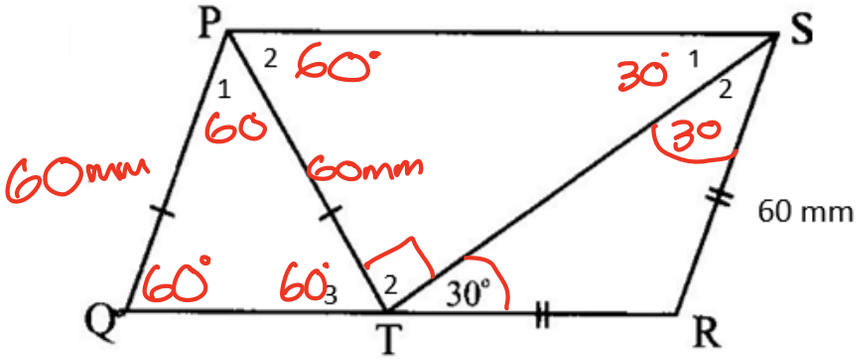
Showing all your calculations, evaluate $\dfrac{\sin 45° \,.\, \sec 30°}{\text{cosec } 60° \,.\, \cos 45°}$. (5)

$$= \frac{\frac{\sqrt{2}}{2} \checkmark \times \frac{2}{\sqrt{3}} \checkmark}{\frac{2}{\sqrt{3}} \checkmark \times \frac{\sqrt{2}}{2} \checkmark}$$

$$= 1 \checkmark$$

QUESTION 7 [8 Marks]

PQRS is a parallelogram. T is a point on QR such that SR = TR = 60 mm and PQ = PT.
$S\hat{T}R = \hat{T}_1 = 30°$.



Calculate the length of QT giving reasons for all statements. (8)

PQRS parm ✓

PQ = SR = 60mm ✓ opp sides parm

$\hat{S}_2 = \hat{T} = 30°$ ✓ (SR) SR = TR

$\hat{R} = 120°$ ✓ (SR) sum $L^s \triangle$

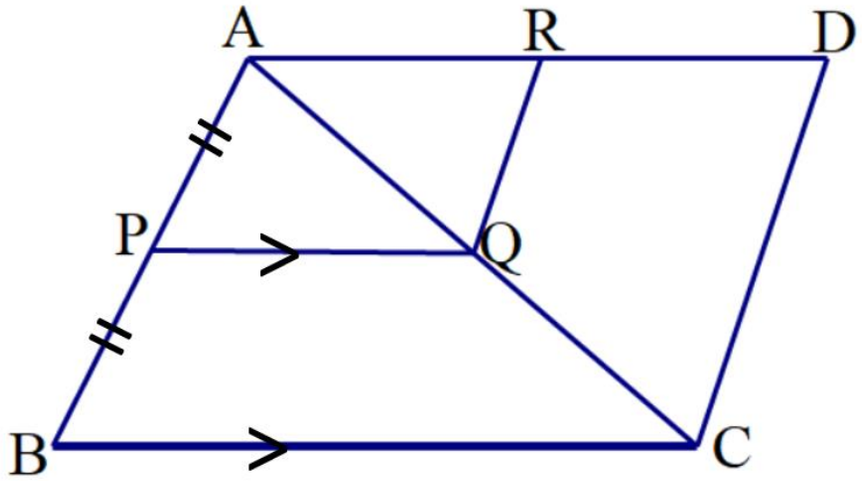
$\hat{Q} = 60°$ ✓ (SR) must include ∥ coint L^s ∥ lines

$T_3 = 60°$ ✓ (SR) PQ = PT

QT = PQ ✓ = 60mm sides of equilat $\triangle$

QUESTION 8 **[8 Marks]**

In the diagram below P is the midpoint of AB and PQ∥BC.



(a) Explain why Q is the midpoint of AC. (3)

AP = PB ✓ PQ || BC ✓

so AQ = QC ✓ line thru midpt || 2nd side

(b) It also given that PARQ is a parallelogram, R is the midpoint of AD and that $BP = 15$ units.

accept converse midpt thm

Calculate, giving reasons, the length of CD. (5)

RQ = AP = 15 ✓ opp sides parm

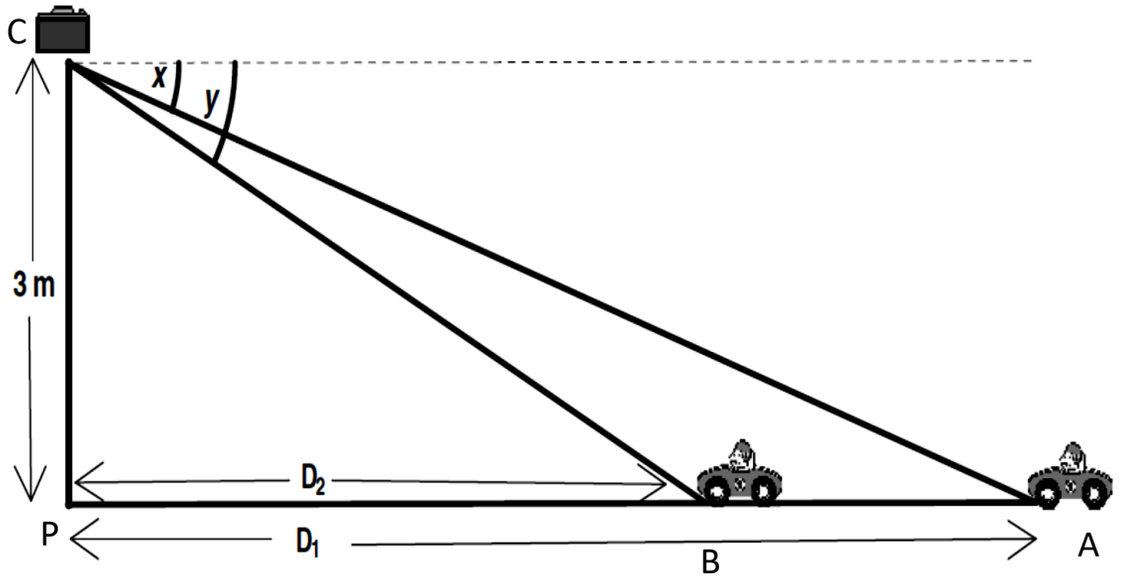
AQ = QC ✓
AR = RD

AR = ½ CD ✓ midpt thm

AD = 30 units ✓

QUESTION 9 [9 Marks]

A laser speed trapping device (C) is mounted on a pole that is 3m high. The device measures the initial angle of depression (angle marked *x*) of a car at position A. It then measures the angle of depression (angle marked y) again 1 second later when the same car reaches position B. These measurements are used to work out how far the car has travelled in 1 second and then to determine whether, or not, the driver is breaking the speed limit.



[Picture taken from Western Cape Exemplar Assessments 2008, Western Cape Education Department, p38]

(a) Show that the initial distance marked AP = D_1, is given by $D_1 = \frac{3}{\tan x}$ and the distance BP, one second later, is given by $D_2 = \frac{3}{\tan y}$. (2)

$\frac{3}{D_1} = \tan x$ ✓ $\therefore D_1 = \frac{3}{\tan x}$

$\frac{3}{D_2} = \tan y$ ✓

$\therefore D_2 = \frac{3}{\tan y}$

(b) If $x = 3{,}5°$ and $y = 6°$, calculate the distance AB (in metres) travelled by the car in 1 second. (4)

$D_1 = \frac{3}{\tan 3{,}5} = 49{,}05$ ✓

$D_2 = \frac{3}{\tan 6} = 28{,}54$ ✓

$AB = 49{,}05 - 28{,}54 = 20{,}51\ m$ ✓ ✓

(c) If the speed limit is 60 km/hour, determine whether, or not, the motorist is exceeding the speed limit.
Show all calculations. (3)

$20{,}51\ m/s \longrightarrow 73{,}84\ km/h$ ✓

$\div 1000 \times 3600$ ✓

∴ Yes motorist is speeding. ✓

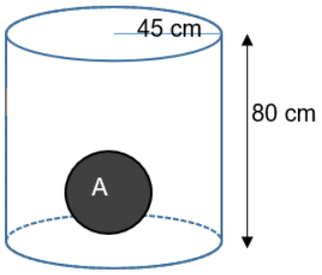
QUESTION 10 [10 Marks]

USEFUL FORMULAE

$Volume\ of\ a\ sphere = \frac{4}{3}\pi r^3$ $Volume\ of\ a\ cone = \frac{1}{3}\pi r^2 h$

$Volume\ of\ a\ cylinder = \pi r^2 h$

The diagram below shows a solid metal sphere, A, with diameter 30cm, placed inside an empty cylindrical tank with radius 45 cm and height 80 cm.
The cylinder is then filled with water.



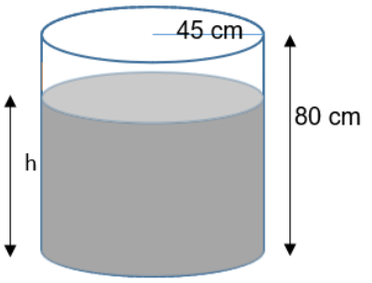
(a) Calculate the volume of the metal sphere. (2)

$$V = \frac{4}{3}\pi (15)^3$$
$$= 4500\pi$$
$$= 14137,2\ cm^3$$

(b) Calculate the volume of water required to fill the tank. (3)

$$V = \pi (45)^2 \times 80 - 14137,2$$
$$= 162000\pi - 4500\pi$$
$$= 157500\pi$$
$$(494\,800,8\ cm^3)$$

(c) The sphere is then carefully removed from the tank.





Calculate h, the height of the water remaining in the tank. (Assume that none of the water is spilt as you remove the sphere.) (3)

$$157\,500\pi = \pi(45)^2 h$$
$$494\,800{,}8 = \pi(45)^2 h$$

$$77{,}8\text{ cm} = h$$

(d) The sphere is then melted down and reformed into a solid cone with height 25 cm. Calculate the radius of this new cone. (2)

$$4500\,\pi = \frac{1}{3}\pi r^2 25$$

$$540 = r^2$$

$$23{,}2 = r$$

TOTAL 100

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